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Reasoning with Data

Statistical thinking recognizes that observations vary. A pattern in collected data may reflect a real process, random variation, measurement error, or selection effects. Good reasoning asks how the data were produced before deciding what a summary or model means.

A population is the group or process of interest; a sample is the portion observed. A parameter describes a population, while a statistic summarizes a sample. The two should not be confused: a sample statistic is an estimate and may change if a different sample is collected.

Key points

- Define the population and observation unit.
- Distinguish sample evidence from population claims.
- Consider how measurement and collection processes affect results.

Summarize Distributions

The mean uses every numeric value but can be strongly affected by extreme observations. The median is the middle value after sorting and is more resistant to extreme values. Quartiles describe the spread of the middle half of observations; range and standard deviation provide other measures of variability.

A summary should match the distribution. A skewed dataset may be better described by a median and interquartile range than by a mean alone. Always inspect the data visually and state the number of observations because a statistic without context can hide important differences.

Key points

- Use mean and standard deviation for reasonably symmetric data when appropriate.
- Use median and quartiles for skewed data or data with influential extremes.
- Report units, group sizes, and relevant missingness.

Sampling and Uncertainty

Different samples from the same population produce different estimates. This sampling variability is expected and can be studied by collecting repeated samples or using statistical models. Larger samples often reduce random uncertainty, but they do not automatically remove bias from a poor sampling frame or flawed measurement.

A confidence interval gives a range of estimates generated by a procedure with a stated long-run coverage property under its assumptions. It is not a guarantee that a particular interval contains the parameter with a specified probability after it has been computed. Explain uncertainty in accessible terms.

Key points

- Check how participants or observations entered the sample.
- Distinguish random error from systematic bias.
- Report uncertainty alongside estimates rather than presenting precision without context.

Hypothesis Tests and Practical Meaning

A hypothesis test compares observed data with a model or null assumption. A p-value describes how unusual data at least as extreme as those observed would be if the null model and its assumptions were correct. It is not the probability that the null hypothesis is true.

A threshold such as 0.05 is a convention, not a boundary between truth and falsehood. Statistical significance does not measure the size or usefulness of an effect. Report effect estimates, uncertainty, study design, and practical context; consider how multiple comparisons increase the chance of apparently notable findings.

Key points

- State the null and alternative questions clearly.
- Do not interpret p-values as effect size or proof.
- Assess practical importance and assumptions in addition to statistical evidence.

Association, Causation, and Study Design

Correlation describes how two variables vary together. It does not, by itself, show that one variable causes the other. A third factor may influence both, the direction may be reversed, or the association may arise through selection or chance. Causal claims require a design and assumptions that address these alternatives.

Randomized experiments can support causal interpretation when implemented well, while observational studies often require careful adjustment and still retain uncertainty. The wording of a conclusion should match the design: describe an association when the evidence only supports association.

Key points

- Draw a simple causal diagram or list plausible confounders.
- Check whether groups are comparable and measurements are consistent.
- Avoid causal wording unless the study design justifies it.

Statistical Investigation

Choose a measurable question and identify the population, outcome, groups, and sampling process. Summarize one numeric variable with a plot and suitable center and spread. If comparing groups, report their sizes and show both the difference and uncertainty rather than relying on a single p-value.

Write a conclusion that separates observation from interpretation. Include the effect estimate, uncertainty, limitations, and at least one alternative explanation. A transparent report should make it possible to understand what was tested and what cannot be concluded.

Key points

- Deliver: question, descriptive summaries, visual comparison, effect estimate, and limitations.
- Review: What does a p-value describe? Why is correlation not causation?
- Check that conclusions address the population and design actually studied.